

MOHAMMAD SAFIEDDINE

Full-Stack Developer · Founder & CEO, H3M Softwares · Applied AI/LLM Engineer

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PROFESSIONAL SUMMARY

Computer Science graduate with production experience shipping full-stack applications (Next.js, FastAPI, PostgreSQL) and applied LLM systems (RAG pipelines, autonomous agents) — including a live document-intelligence API serving real users, a self-founded software company shipping a commercial point-of-sale and e-commerce platform, and a remote internship with a US-based company shipping customer-facing features. Seeking a full-stack or applied AI engineering role where I can bring end-to-end ownership and production-minded engineering to a product team.

EDUCATION

M1 – Master's Degree in Computer Science

2025 – 2026

Lebanese University, Faculty of Science 5 — Nabatiyeh, Lebanon

B.S. in Computer Science

Sep 2021 – Jun 2024

Lebanese University, Faculty of Science 5 — Nabatiyeh, Lebanon

Relevant Coursework: Algorithms & Data Structures, Software Engineering, Database Management Systems, Operating Systems

EXPERIENCE

Founder & CEO | H3M Softwares

2025 – Present

Remote · h3msoftwares@gmail.com

- **Founded and lead a software company** building commercial retail products, including **H3M POS** (multi-warehouse POS/back-office platform, ~46k LOC) and **Ali's Store** (bilingual EN/AR e-commerce platform), from architecture through delivery.
- Own full-stack architecture, security design, and release engineering across both products, spanning Next.js/React frontends, Node/Express/Prisma backends, and — for the desktop product — an Electron packaging and licensing layer.
- Operate as sole/lead engineer across the full product lifecycle: requirements, data modeling, implementation, automated testing (Vitest, Playwright, k6), and client delivery.

Software Developer Intern | Helpa

Apr 2026 – Jun 2026

Texas, United States (Remote) · Fixed-term internship

- Shipped 4 production features to a live customer-facing Next.js/PostgreSQL/Node.js application, spanning both frontend UI and backend API layers.
- Identified and resolved 3 database performance bottlenecks through query optimization, and fixed 3 production bugs on customer-facing applications.
- Recognized as a top-performing intern for code quality and proactive delivery, based on manager feedback.

Freelance Web Developer | Self-Employed

Jul 2024 – Present

Remote

- Delivered 3+ custom web applications for independent clients, cutting manual process time by an estimated 30% through workflow automation.
- Owned each engagement end-to-end — requirements gathering, UI/UX design, development, and post-launch support — translating client needs into responsive, accessible interfaces.
- Core stack: React.js, JavaScript, HTML/CSS, Express.js, Electron.js, MySQL.

PROJECTS

H3M POS — Retail Point-of-Sale & Back-Office Platform

2025 – 2026

Next.js, React, Node/Express, TypeScript, Prisma, PostgreSQL, Electron, Redux Toolkit, TanStack Query · H3M Softwares

- Built a full POS and back-office system — checkout, multi-warehouse inventory, purchasing, supplier/customer ledgers, shifts, and analytics — shipping both as a containerized web app and as a single self-contained Windows desktop installer bundling its own PostgreSQL database, API, and licensing.
- Implemented concurrency-safe checkout with idempotency-key replay protection and row-level locking so concurrent registers can't oversell stock or double-charge on retry.
- Designed a signed, machine-bound Ed25519 licensing system with roll-back-resistant expiry and a code-integrity manifest, built to remain enforceable after JavaScript obfuscation.
- Delivered full bilingual EN/AR support with true RTL across ~1,960 translation keys per language, and a 32-permission RBAC system with an audit log of every privileged action.

- Scale: ~46,000 LOC across ~50 Prisma models, 38 App Router screens, and ~160 automated tests (Vitest, Playwright, k6 load testing).

Ali's Store — Bilingual E-Commerce Platform

2026

Next.js, React, Express, Prisma, PostgreSQL, React Query, Redux Toolkit

- Built a production storefront and admin back office for a retail client — bilingual EN/AR with full RTL, cash-on-delivery checkout, and a granular role-based admin panel.
- Implemented security-first auth: JWT access tokens with rotating refresh tokens and reuse detection, Argon2 hashing, step-up re-authentication, and account-enumeration-resistant responses.
- Built a COD checkout flow with email-OTP verification, live coupon/discount resolution, per-governorate delivery fees, and an advisory lock to serialize a shopper's concurrent checkouts.
- Shipped an owner-controlled home page (banners, featured collections, smart product rows) and a typo-tolerant catalog search backed by Postgres pg_trgm, validated with Vitest integration tests, Testing-Library, and a Playwright E2E sweep.

Document Assistant API — RAG Document Intelligence Platform

2025

Python, FastAPI, ChromaDB, Sentence Transformers, Groq (Llama 3.3 70B), React.js, Docker

- Built and deployed a production REST API enabling users to upload documents (PDF, DOCX, PPTX, TXT) and query them through a RAG pipeline — live on HuggingFace Spaces (backend) and Vercel (frontend).
- Implemented per-user ChromaDB collections and isolated file storage for full session isolation, with 24-hour auto-cleanup and rate limiting.

Smoke Dash — Full-Stack Admin & Audit Platform

2025

Next.js, TypeScript, Fastify, PostgreSQL, Prisma, Zustand, React Query, JWT, Zod

- Architected a role-based (RBAC) admin platform with full audit trails — IP address, timestamp, and before/after value diffs — for every CREATE/UPDATE/DELETE action.
- Used a Route → Controller → Service → Prisma → DB architecture with Zod validation and transaction-safe Prisma migrations.
- Supports 3 concurrent roles/permissions across 5 managed entities.

Agentic RAG — Football Intelligence System

2025

Python, FastAPI, ChromaDB, React.js, Groq, DuckDuckGo Search

- Built a full-stack AI application combining a RAG pipeline (Wikipedia scraping, chunking, Sentence Transformer embeddings, ChromaDB storage) with an autonomous tool-calling agent (document search, web search, calculator, date retrieval).
- Built an automated LLM-as-judge evaluation suite; achieved 87% answer correctness and cut hallucination rate by 93%.

Boss Burger — Restaurant Ordering & Admin System

Dec 2024 – Feb 2025

React.js, Electron.js, Firebase

- Built a customer-facing ordering website paired with a real-time admin dashboard for live order and operations management.

Other Notable Projects

- Smoke POS — offline-capable desktop POS with barcode scanning and encrypted auth (React.js, Electron.js, Express.js, MariaDB); CIS — invoicing, payments, and inventory platform for a client's supplier operations; PTS — clinical patient-tracking tool for visit, medication, and exam history.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL, PHP, Java

Frontend: React.js, Next.js, Redux.js, Zustand, React Query

Backend: Node.js/Express.js, Fastify, FastAPI, Prisma

AI / LLM: RAG Pipelines, AI Agents, LangChain, ChromaDB, Sentence Transformers, Prompt Engineering, LLM APIs (Groq, HuggingFace)

Databases: PostgreSQL, MySQL, MariaDB, Firebase

Infra / Tools: Docker, Git/GitHub, Vercel, Render, HuggingFace Spaces

Languages Spoken: English — Fluent Arabic — Native